Topic: Log-concave density estimation and two-sample testing

Theoretical and statistical framework

Log-concave density estimation is a nonparametric method based on shape constraints, where the underlying probability density function f is assumed to have a concave logarithm. This assumption defines a broad and flexible class of distributions, including Gaussian, exponential and gamma families, while excluding highly irregular or strongly multimodal densities. The log-concavity constraint regularises the estimation problem without imposing a fully parametric specification.

Given an independent sample $X_1, \ldots, X_n$ from an unknown distribution, the log-concave density estimator is defined as the nonparametric maximum likelihood estimator under the concavity constraint on $\log f$. Unlike kernel density estimation, this approach does not require the selection of a smoothing parameter, thereby avoiding bandwidth sensitivity and subjective tuning. The resulting estimator is piecewise exponential, automatically adapts to the scale of the data, and exhibits well-behaved tail behaviour.

From a statistical inference standpoint, log-concave density estimation provides a principled compromise between parametric efficiency and nonparametric robustness. The likelihood-based nature of the estimator allows hypothesis testing procedures to be constructed directly from likelihood ratios, while remaining robust to moderate deviations from classical distributional assumptions.

In the two-sample setting, the objective is to test

$$H_0 : F_1 = F_2$$

against the alternative that the two distributions differ, under the assumption that both densities are log-concave. Under the null hypothesis, both samples are assumed to arise from a common log-concave density, whereas under the alternative each sample is allowed its own log-concave estimator. Comparing the maximised log-likelihoods under these two scenarios yields a natural likelihood ratio statistic. Since the finite-sample distribution of this statistic is analytically intractable, inference is typically conducted using permutation methods.

This framework enables the assessment of distributional differences beyond simple location shifts, capturing changes in dispersion and overall shape while avoiding parametric misspecification. Classical nonparametric tests, such as the Kolmogorov–Smirnov and Wilcoxon tests, are considered as benchmarks for comparison.

Data analysis

The analysis is based on the `diamonds` dataset from the `ggplot2` package, which contains prices and characteristics of over 50,000 diamonds. To ensure comparability and interpretability, the response variable considered is the logarithm of the diamond price. Two large subpopulations are defined according to cut quality: *Ideal* ($n_1 = 21{,}551$) and *Premium* ($n_2 = 13{,}791$).

The mean log-price for Ideal diamonds is 7.639, compared to 7.951 for Premium diamonds, indicating a clear difference in central tendency on the logarithmic scale. Exploratory graphical analysis reveals a systematic rightward shift of the Premium price distribution, together with mild differences in dispersion and shape.

Formal two-sample testing provides overwhelming evidence against the null hypothesis of equal distributions. Both the Kolmogorov–Smirnov test and the Wilcoxon rank-sum test yield p-values smaller than 2.2×10^{-16}. Given the very large sample sizes involved, such extreme p-values are expected even under moderate distributional differences and should therefore be interpreted in conjunction with effect size and graphical diagnostics rather than in isolation.

Log-concave density estimators are fitted separately to each group and compared with kernel density estimators. Kernel-based estimates exhibit local irregularities driven by bandwidth selection, whereas the log-concave estimators provide smooth, shape-constrained representations that emphasise global distributional features. The fitted densities suggest not only a location shift, but also differences in dispersion and overall structure between the two groups.

From a practical perspective, the difference in mean log-prices corresponds to a multiplicative price premium on the original scale, indicating that the observed statistical differences translate into economically meaningful effects. From an inferential standpoint, rejecting the null hypothesis does not imply that the two groups differ at every price level, but rather that their overall distributional structures are incompatible with a common generating mechanism under the log-concave assumption. This highlights the advantage of distributional testing over purely mean-based comparisons, particularly in large samples where subtle but systematic differences can be detected.

Figures and references

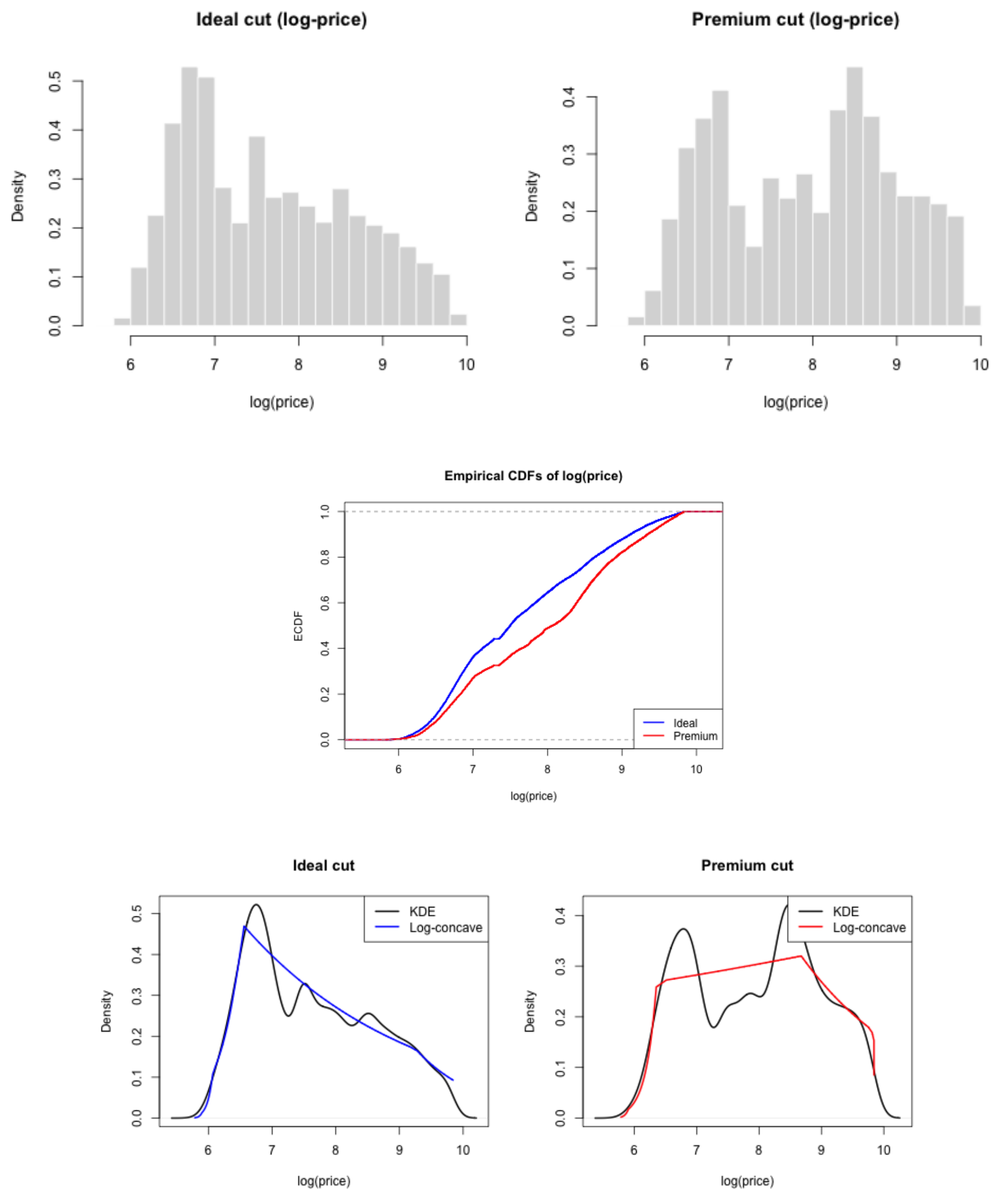

Figure 1: Histograms of log-price for Ideal and Premium cuts (top), empirical cumulative distribution functions (middle), and kernel density estimates compared with log-concave density estimators (bottom).

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2. Wasserman, L. (2006). *All of Nonparametric Statistics.* Springer.

3. Agresti, A. (2013). *Categorical Data Analysis.* Wiley.